41097 — Software Engineering Studio 3A

Reflective Ability and Design Thinking

Assessment 1-2

Sprint Weekly Journals

Journal 1 – 15.03.2021

Stand-up meeting:

1. Things I have done since last week's meeting? - I did some research on different technologies (unity, xl) that can be used in our project - Did some target market research and the risks involved in the overall project. - I collaborated with the team to work on the Team Charter.
2. Things I am going to get done in the following week? - Work on the team with setting up the back end for our project - Start learning more about the XR Designs
3. What obstacles/blockers I have? - Lack of knowledge in unity and about XR Designs - Inexperience with Unity based implementations.
4. What can I get help with and from whom? (matching the above answers if you have) - For me researching on my own on different portal such as Stack overflow and Google has worked the best in the past studios and something I would like to continue this semester too. - Seek help from more experienced team members if needed to.

Retrospective:

1. Acknowledgments • Team Meetings are really useful to clarify all the concerns and to ensure everyone in the team is on the same page • Working alongside the team to share all the research and any interesting findings.
2. What went well or very well in this sprint? • The completion of set individual task by the team members • Good initial clarity on the assigned tasks to each individual and the their requirements.
3. What didn't go so well in this sprint? • Not much as it was the first week of the project commencement.
4. Next actions (specific tasks, assigned to specific person/people, arrangement, management) • Start working on the backend • Start looking at the different user stories and their implementation.

Journal 2 – 22.03.2021

Stand-up meeting:

4. Things I have done since last week's meeting? - I created a MongoDB account for myself. - I did some research on MongoDb and completed some online video tutorials for my knowledge as I have never used Mongodb before. - Worked with team mates on connecting Mongodb to our project
5. Things I am going to get done in the following week? - Start completing my tasks on CRUD operation for (training) Courses - Work on files which needs to be added in the backends. -
6. What obstacles/blockers I have? - Lack of knowledge in MongoDB and about XR Designs - Inexperience with Unity based implementations.
4. What can I get help with and from whom? (matching the above answers if you have) - Seek help from more experienced team members in our backend team and learn from other projects featuring Mongodb backend.

Retrospective:

1. Acknowledgments • Team Meetings are really useful to clarify all the concerns and to ensure everyone in the team is on the same page • Working alongside the team to share all the research and any interesting findings.
2. What went well or very well in this sprint? • Working on the backend smoothly • Learning on different mongodb implementations.
3. What didn't go so well in this sprint? • Couldn't focus too much on my other tasks of being the documentation lead
4. Next actions (specific tasks, assigned to specific person/people, arrangement, management) Work on my assigned tasks for this week Work on user authentications in web applications

Journal 3 – 29.03.2021

Stand-up meeting:

7. Things I have done since last week's meeting? - I started working on the CRUD operation for courses. - Worked with team mates on fixing backend issue regarding appsettings.json and gitignore file.
8. Things I am going to get done in the following week? - Complete any further task on CRUD operation for Courses and have it reviewed by stream lead for any feedback. - Work on recreating existing endpoints in expressJS.
9. What obstacles/blockers I have? - Lack of experience with MongoDB - Not enough knowledge about XR Designs - Inexperience with Unity based implementations.
4. What can I get help with and from whom? (matching the above answers if you have) - Research on Google + youtube for any issues/challenges that I face with implementations. - Seek help from more experienced team members in our backend team.

Retrospective:

1. Acknowledgments • Team Meetings are really useful to clarify all the concerns and to ensure everyone in the team is on the same page • Working alongside the team to share all the research and any interesting findings.
2. What went well or very well in this sprint? • Backend was made operational.
3. What didn't go so well in this sprint? • Couldn't focus too much on my other tasks of being the documentation lead
4. Next actions (specific tasks, assigned to specific person/people, arrangement, management) Work on user authentications in web applications Start completing any pending documentation tasks for the previous weeks.

Journal 4 – 12.04.2021

Group Name: 1 **# of Spring and Date**: Journal 4- 12/4/2021

Your Name: Lakshya Sharma **Student ID**: 13229911

Stand-up meeting:

10. Things I have done since last week's meeting? - I created the basic endpoints for ingredients. - I did some training on the basics of ExpressJS
11. Things I am going to get done in the following week? - Work on any errors that come up in the existing endpoints. - Work with the front end team in assisting them with connections.
12. What obstacles/blockers I have? - Lack of experience with connection with front end and back end. - We used Express JS instead of Typescript
4. What can I get help with and from whom? (matching the above answers if you have) - Research on Google + Youtube for any issues/challenges that I face with connections. - Seek help from more experienced team members in our backend team.

Additional Notes:

GitHub Commits: `cee45ca396d05980aa3ca2b4b060c7b64363eff4`

SES 3A Design Thinking

Empathise (scope)

The core fundamantals of Virtual Organizational Training Experience is to focus on its target customer of large business and organizations that are willing to advance their training experiences using the most recent technology. The target customers would like a solution such as a platform with training experience that would allow them to create their own custom XR training modules. Due to time constraints, our group has planned to provide the solution to clients request within their requested time frame. In the initial stage, our group will focus mainly on finalizing the web application component of the product, and one particular training scenario to give client a full walkthrough of how the final software/platform would like to go from beginning to end to reach the solution in place.

Our target customer would reach their endgame smoothly due to our software 's capability. In the long term, we plan to make our platform more readily accessible to everyday consumer who will be able to use the VR/AR hardware by making it more affordable.

Our product would be the preffered option by our target customer due to following reasons:

- Our training models will be easily recognizable and match industry best practices.
- They can be used to train across any industry.
- There would be set constraints in place to decrease any of the health risks such as eye, neck, or arm fatigue which our competitors are not providing.

Finally our software would allow our target customer to achieve their endpoint by focusing on their pain points such as reducing unnecessary financial expenses, increasing the overall effectiveness of their training procedures through the use of VR/AR verses in the real life.

Define (make reasonable assumptions)

Our goal was to ensure that the backend database was recording the correct details. For example, supervisor can create courses for the trainees they complete the task. I completed and setup the ingredients arm of our database.

User Stories:

1. Supervisor wants to gather all the different ingredients so they can use it to create a reciepe.
2. Supervisor wants to pick a specific ingredient from the database.
3. Supervisor wants to create a new ingredient.
4. Supervisor wants to update the ingredient.
5. Supervisor wants to delete the ingredient.

Ideate (User Story and/or User Story Map)

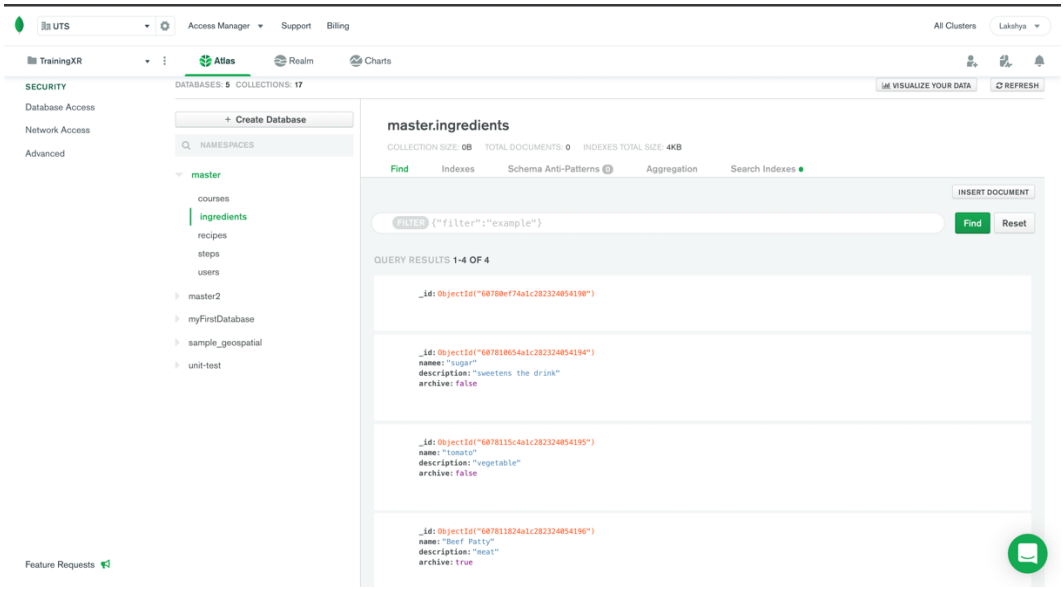
User Story Map

We had two proposed solutions:

1. **MongoDB – NO SQL with asp.net framework** – We wanted to use this spefic framework at the start of the sprint because its directly coded into csharp which is being used by our unity team in the group as well. This would have allowed our back end team to help the unity team down the line because of the same language and overall easier transition.

2. **MogoDB – NO SQL with node js/express js** – We decided to use framework at the end because asp.net didn't have any authentication libraries, so we would have to create it ourselves. Our other focus was to make the project effecient and to overall reduce the time of deployment, hence we decided to used express js. Majority of the people in our group also had had experience from previous studios with this framework and therefore were most comfortable with in implemenatation.

Prototype



Database on MogoDb

Above photo depicts our Mongo database/backend for the project. There are 5 core databses including recipes, courses, steps and ingredients and user. I completed the Ingredients database which is fully functional. The Ingredients tab can save the name of the item, the description related to that product and has an option to be archived if needed to by the client.

We had decided to keep and store ingredients in a separate database. This would lead to each ingredient to have their own unique Code in database, which can then be used in the recipes instead of the actual name of ingredient. This was done to ensure there is clarity amongst the user and to expand on the efficiency for the client when making recipes.

Test

```
107    ###
       Send Request
108    GET http://localhost:4000/ingredient
109
110
```

REST Client

This form of testing is called REST Client, which is a VS Code extension. REST Client is a process to summon a REST service API which is then exposed for communication by any system or service provider. We put 3 hashtags to initialise the test, we then put get request and then added the link with the ingredients router (/ingredient) after the http://localhost:4000. We did the following testing method to gather all the ingredients. REST stands for "Representational State Transfer," which is a new way of communication between any two systems at a given point of time.

Analysis

Initially our team plan was to implement the database through asp.net framework but due to all the limitations discussed earlier and we decided to move to the plan was express node.js

Our group gave important to test the interface with users to ensure that the our frameworks simple to use and highly efficient. Our group across all the streams made sure that the right values are shown/hidden, make sure the frontend data matches the corresponding fields in the database. If incorrect data was to be shown in other way, it would be a violation of user privacy.

Having high levels of security in pace for our project was a big priority for our team and there for we decided to go for Bearer Tokens. A Bearer Token is an opaque string, not intended to have any meaning to clients using it. Bearer Tokens are usually the predominant type of access token used with OAuth 2.0. Our goal was to ensure that to access any kind of data in our database that right credentials were linked and then used for the access.